Glucose Digital Twin: architecture

T2D / prediabetes post-meal excursion twin. Mechanistic core + UKF state estimation + boosted ML residual. Decision support for lifestyle, reviewed by a clinician.

1 DATA IN

CGMacros (PhysioNet)
Dexcom G6 + Libre, meal macros,
Fitbit METs, HbA1c. 29 pts, HbA1c >= 5.7
Development + training (CC BY-NC-SA)

ShanghaiT2DM (figshare)
15-min Libre, free-text diet logs
38 non-insulin T2D patients
EXTERNAL validation only (CC BY 4.0)

Synthetic generator
Structurally different from the twin
Stress tests + counterfactual check
Never a headline number

Static / clinical context
HbA1c, age, BMI (prior on insulin sensitivity); Indian meal library (18 items, carbs + slow fraction)

Interoperability (mock / roadmap)
FHIR R4 Bundle out (LOINC 4548-4, 2339-0, ICD-10 E11). ABHA / ABDM consented pull: architecture only, no real PHI

FHIR in/out

2 INGESTION (twin/data.py)

Align to a 5-min grid | 15-min Libre linearly interpolated (gaps > 45 min stay missing) | meal text/macros to carbs + slow-absorption fraction

Activity: METs to steps/min proxy (CGMacros) or none (Shanghai). Future meal logs are never used as features (leakage guard).

3 TWIN ENGINE (twin/model.py, twin/pipeline.py)

A Calibrate (per patient)
Bergman-style minimal model + 2-pool gut
State [G, X, F1, F2, S1, S2]
Fit SI, SG, k on 4-h post-meal segments
Empirical-Bayes prior (cohort + HbA1c)
Posterior covariance kept

theta

B Assimilate (online)
Unscented Kalman filter, 1 step / reading
Meal log = known input + 25% size uncertainty. Fills sensor gaps; the twin state updates at every sample

state

C Forecast + risk
ODE roll-out 30 / 60 / 120 min
+ gradient-boosted residual (patient-level CV, pooled across patients)
P(glucose > 180 within 2 h) with empirical per-horizon uncertainty

P

D Alert policy
One pooled threshold, set on calibration windows for
<= 2 false alerts / patient-day
Per-excursion lead time

E What-if simulator
Meal swap / post-meal walk are ODE inputs, re-simulated with 60 posterior samples + portion noise: bands, not point guesses

refit

F Drift monitor and recalibration loop (bidirectional)
24-h mean of the 60-min forecast error vs calibration window;
shift > 20 mg/dL triggers a refit on the last 3 days, prior = current parameters. Closed loop: recommend, simulated / observed response, compare forecast vs outcome, recalibrate. Counterfactual check on synthetic truth: walk effect direction right, size over-stated.

G Evaluation harness (patient-level, time-ordered)
Baselines: persistence, linear extrapolation, ODE+UKF only, meal-rule. Metrics: RMSE/MAE at 30/60/120 min, Clarke error grid, AUROC / AUPRC, Brier + reliability, alerts per day, sensitivity, lead time, drift ablation.
Internal: CGMacros, 5-fold by patient. External: Shanghai with the CGMacros-trained pieces frozen (Libre-15-min-trained residual, no steps / HbA1c features); only per-patient ODE parameters are fitted.
Calibrate on first 3-4 days, test on the rest. Synthetic data is never the headline.

4 SERVING AND CLINICIAN UI

Stdlib HTTP server (twin/server.py)
GET /api/results precomputed twin + metrics
POST /api/whatif live ODE re-simulation
GET /api/fhir/<id> FHIR Bundle
Static: dashboard, three.js (vendored)

Dashboard (web/index.html, web/human.js)
Triage list ranked by 2-h risk | Forecast cone with the real outcome revealed in replay | Risk, alert feed, mechanistic drivers of the 60-min forecast
What-if: Indian meal A vs B (+ walk) with uncertainty bands | Twin state: SI / SG / k with 95% CI + drift events | AGP + time-in-range vs consensus targets
Validation tab: baselines, external Shanghai table, alert operating curve, reliability, per-patient table | 'Twin vs classifier' checklist
3D physical twin: clothed figure walks / eats / sleeps from logged activity; CGM patch and floor ring show glucose zone; X-ray mode shows organs driven by model state (stomach = carbs on board, pancreas = insulin action). WebGL failure falls back to a 2D figure.

Governance Intended use: lifestyle decision support reviewed by a clinician; no diagnosis, no insulin dosing. CDSCO draft MDSW (Oct 2025): plausibly Class B (hypothesis).
DPDP: public / synthetic data only here. Production: consent notice, ABDM consent artefact, store twin parameters not raw streams, erase on withdrawal. No Indian CGM data exists in this work; India-specific priors are labelled assumptions.